neutron stars can have a maximum mass by a crude calculation, just a few pages long. The result was intriguing: He found a maximum mass of 6 Suns for any neutron star. If a detailed calculation gave the same result, then Oppenheimer could conclude that black holes might form when stars heavier than 6 Suns die.

A "detailed calculation" meant selecting a mass for a hypothetical neutron star, then asking whether, for that mass, neutron pressure inside the star can balance gravity. If the balance can be achieved, then neutron stars can have that mass. It would be necessary to choose one mass after another, and for each ask about the balance between pressure and gravity. This enterprise is harder than it might sound, because pressure and gravity must balance each other *everywhere* inside the star. However, it was an enterprise that had been pursued once before, by Chandrasekhar, in his analysis of white dwarfs (the analysis performed using Arthur Eddington's Braunschweiger calculator, with Eddington looking over Chandrasekhar's shoulder; Chapter 4).

Oppenheimer could pattern his neutron-star calculations after Chandrasekhar's white-dwarf calculations, but only after making two crucial changes: First, in a white dwarf the pressure is produced by electrons, and in a neutron star by neutrons, so the *equation of state* (the relation between pressure and density) will be different. Second, in a white dwarf, gravity is weak enough that it can be described equally well by Newton's laws or by Einstein's general relativity; the two descriptions will give almost precisely the same predictions, so Chandrasekhar chose the simpler description, Newton's. By contrast, in a neutron star, with its much smaller circumference, gravity is so strong that using Newton's laws might cause serious errors, so Oppenheimer would have to describe gravity by Einstein's general relativistic laws.* Aside from these two changes—a new equation of state (neutron pressure instead of electron) and a new description of gravity (Einstein's instead of Newton's)—Oppenheimer's calculation would be the same as Chandrasekhar's.

Having gotten this far, Oppenheimer was ready to turn the details of the calculation over to a student. He chose George Volkoff, a young man from Vancouver, who had emigrated from Russia in 1924.

Oppenheimer explained the problem to Volkoff and told him that the mathematical description of gravity that he would need was in a textbook that Richard Tolman had written, *Relativity, Thermodynamics, and*